

To solve this, I work with four principles that help create clarity: Define, Target, Scale, and Grow.

Define

The first principle is to define the level of intelligence actually required in a solution. We must be deliberate here because not every problem needs an LLM or a deep learning model. The principle can be handled on three levels.

Good old-fashioned AI (GOFAI):

At its core, GOFAI is a rule-based system. The logic sits in the code. If the business rule changes, we simply edit the rule. GOFAI remains extremely useful, practical, and the right answer for many use cases. There is no need to complicate everything.

Machine learning: We use ML when the system needs to learn from patterns rather than rules. This is where training data, predictions, supervised and unsupervised learning come in.

Complex AI: This includes deep learning, dynamic models, advanced architectures, and LLMs. While AI needs to evolve, setting limits is just as important. At each stage, simply keep evaluating the question: 'Is the current level of complexity strictly necessary to resolve the customer's problem? Often, the honest answer is 'No.

Developers must therefore pause and choose the minimum intelligence level that meets the requirement. Nothing more. Nothing less.

Target

The second principle is about targeting the business problem the right way. AI application development looks similar to the traditional SDLC (software development life cycle), but there is one major difference: refinement.

In software development, we design, code, test, deploy, and maintain. But AI adds a permanent, mandatory last step: refinement.. This is not the same as iterative development. Refinement is a continuous loop built into the system

itself. Feedback from customers and users must be collected, analysed, and fed back into the model regularly. Without refinement, an AI product decays quickly.

This means two things:

1. A feedback mechanism is a non-functional requirement. It must exist either inside the code or as a direct customer touchpoint.
2. Developers need a top-down view—from business case to deployment—because AI is no longer an isolated IT delivery. It affects the business model, not just the software.

In earlier years, we did not revisit the business case frequently. In AI, we must. Needs evolve. Models drift. Use cases change direction. Developers can no longer focus solely on implementation; we need to understand the charter, the model, and the end-to-end refinement cycle.

Scale

The third principle is about scaling. Traditionally, we designed modules based on functions: Module 1 performs Function A; Module 2 performs Function B. Testing followed the same pattern.

But AI development today requires a shift from functional thinking to service-level development.

Why?

Because every module now calls external services—LLMs, APIs, cloud platforms—not once, but many times in a single workflow. We no longer live in a world where only one or two modules connect externally. Now, every module does.

This is where microservices become essential. If your organisation uses GPT today and switches to another provider next year, how will your system adapt? If a model upgrades, will you redo your entire codebase?

With microservices, a change in one service can flow across the organisation without rewriting everything. This is the power of thinking in terms of services rather than functions.

Cloud providers have also evolved. Earlier, we worked with infrastructure-as-a-service. Today, we rely heavily on platform-as-a-service model APIs, ML services, and LLM endpoints, so our architecture must match the service mindset.

Grow

The fourth principle is growth. To move forward responsibly, we must understand where we stand. Gartner's AI maturity model describes five levels:

1. **Awareness:** Employees understand AI basics.
2. **Active:** Teams experiment with POCs, hackathons and low-hanging use cases.
3. **Operational:** AI improves efficiency through internal operations. Most organisations today are here.
4. **Product:** AI powers products delivered to customers, with confidence in accuracy and reliability.
5. **Sentinel:** Fully autonomous, zero-human-touch decision-making systems. Autonomous driving is an example.

Importantly, organisations do not move through these levels sequentially. You cannot finish Level 1 and then begin Level 2. Instead, activities across all levels must run in parallel.

For instance, a company can be running an operational AI system while parallelly creating awareness programmes for responsible AI. Growth is not linear; it is layered.

Awareness increases with company-wide training programmes, expert lectures, and AI meets. Active exploration comes from hackathons and quick POCs (proofs of concepts). Operational impact usually comes through central teams or centres of excellence that identify use cases and implement them. Product-level confidence comes with stronger computing, streamlined pipelines, and model governance. Sentinel level is the long-term vision and represents the highest form of autonomous intelligence.

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